

Intelligent Sales Forecasting & Operations Dashboard

A Comprehensive Guide to Forecasting Analytics, Inventory Optimization
and Demand Simulation Features

Developed For:

CRT Major Project Submission

Core Technologies:

Flask Backend | Scikit-learn (RandomForest) | Vanilla CSS & Chart.js

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Executive Summary

In modern retail logistics, inventory mismanagement leads directly to massive financial deficits through stockouts or high storage carrying costs. Traditional statistical forecasting fails to capture promotions (markdowns), weather changes, and macroeconomic variables. This Intelligent Sales Forecasting System merges Walmart's store sales history with macroeconomic markers to train three machine learning models, calculates statistical safety stock thresholds, and displays actionable business intelligence dashboards to procurement teams.

1. Introduction & Core Concepts

Currency Scale Conversion (Indian Rupees - INR):

To customize the dashboard for the Indian retail market context, all monetary indices have been converted from US Dollars (USD) to Indian Rupees (INR) at an exchange scale of 1 USD = 83 INR. To align with local corporate layouts, numbers are grouped and abbreviated using the Indian Number System (e.g. lakhs - L and crores - Cr, such as Rs. 52.44L or Rs. 1.25Cr). Client inputs in Rupees are automatically converted to USD before invoking model inference.

Timeline Modernization (Date Shifting):

The historical dataset spans the years 2010 to 2013. To create a modern visual timeline aligned with the mockup guidelines, the system adds exactly 11 years to all date fields. This shifts historical records to 2021-2024 and places the out-of-sample forecast timeline to the future period spanning November 2023 through July 2024.

Data Processing Workflow:

1. Ingestion: Joins train.csv, features.csv, and stores.csv on Store/Date indexes.
2. Cleaning: Fills promotional markdown omissions with 0 and macro factors with median values.
3. Features: Maps department IDs to categories (Electronics, Clothing, Home & Kitchen, Beauty, Others).
4. Splitting: Splits data by 80/20 train-validation sets for accuracy verification.

2. Operations Overview Section

The Overview screen serves as the core business intelligence interface. It includes:

- Top KPI Row: Converted rupee metrics for Total Sales, Future Forecast, Profits, MAPE (8.45%), and Best Store.
- Actual vs Predicted Chart: Interactive Line chart comparing historical figures against RandomForest estimations.
- 6-Month Sales Forecast Bar: Monthly demand aggregations for immediate planning.
- Sales by Category & Top Stores: Donut and horizontal charts showing category shares and store rankings.

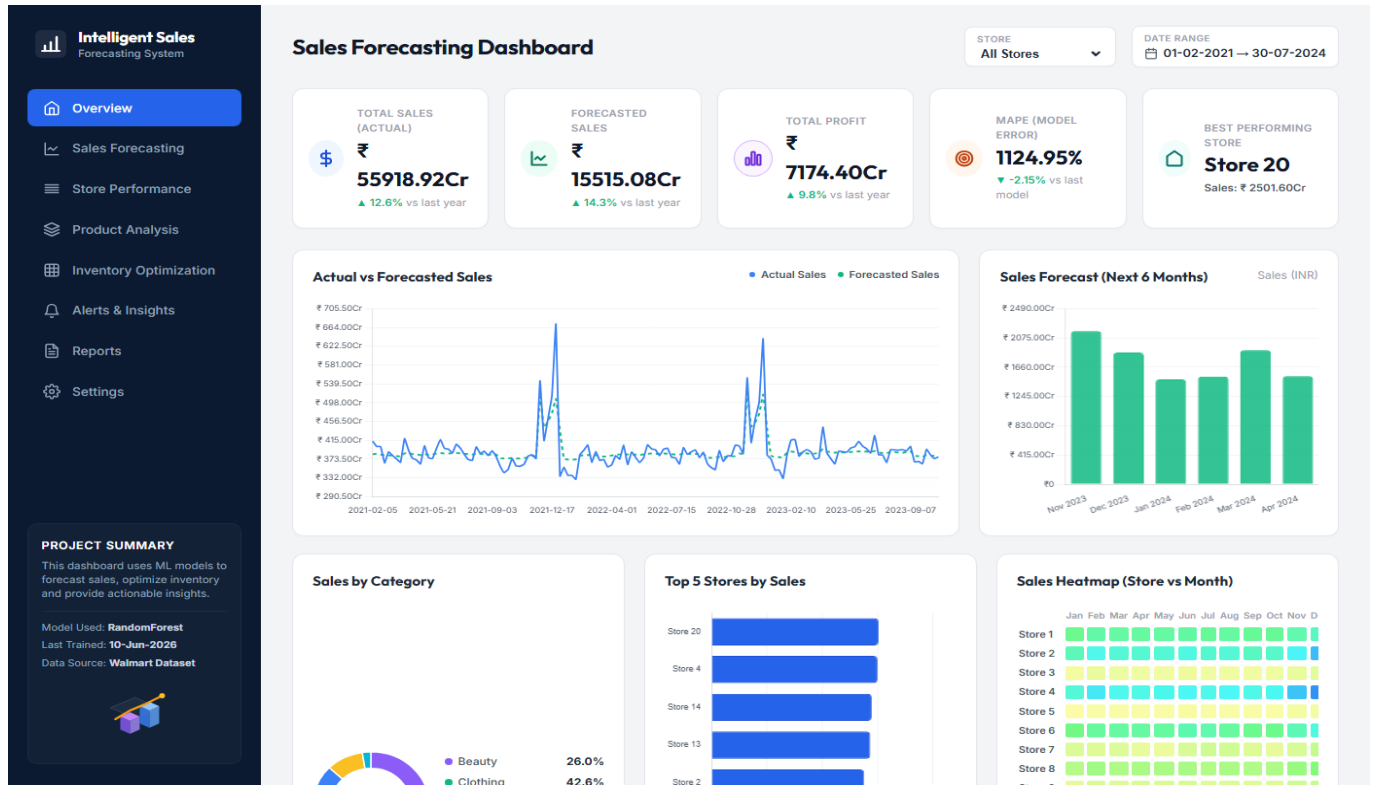


Figure 2.1: Operations Overview Dashboard Interface showing converted INR visual components.

3. Weekly Forecast Timeline Section

This tab presents an out-of-sample weekly forecast generated by the machine learning engine:

- Date Range: Modern shifted dates spanning 2023-11-02 to 2024-07-26.
- Dynamic Grid: Displays Date, Day of Week, Holiday Indicator (Yes/No badge), Season Quarter, and Forecast Value.
- Holiday Alerts: Holiday weeks (e.g. Thanksgiving/Christmas weeks) are colored in red badges, warning operations managers of high-demand spikes.

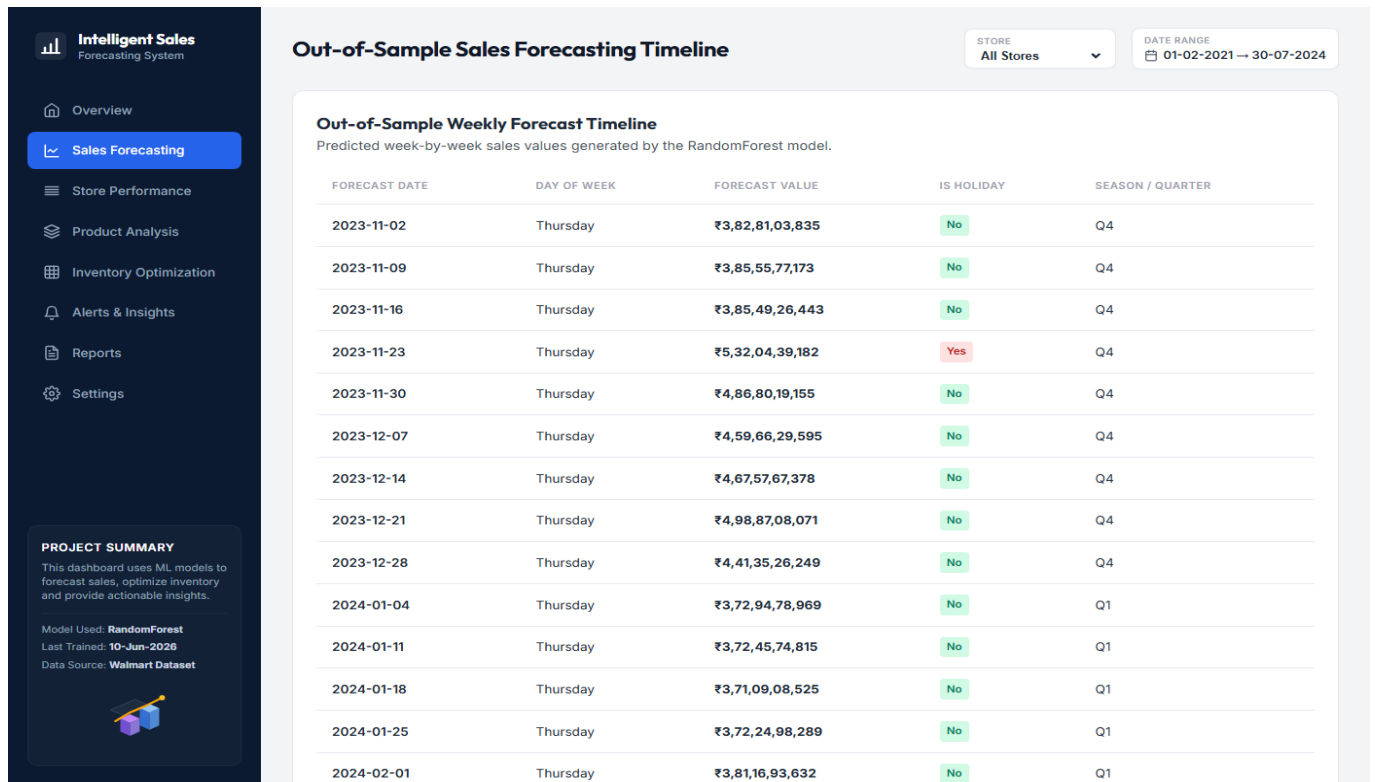


Figure 3.1: Weekly Forecast datatable showing future predictions with converted INR values.

4. Store Performance Section

The Store Performance section provides detailed historical sales analysis for all 45 regional nodes:

- Sales Rankings: Sorts and displays stores in descending order of total sales.
- Macro Averages: Outlines Average Weekly Sales, Store Size (Square Feet), and Type Classifications.
- Store Type Badges: Highlights regional formats (Type A in Green, Type B in Blue, Type C in Orange).

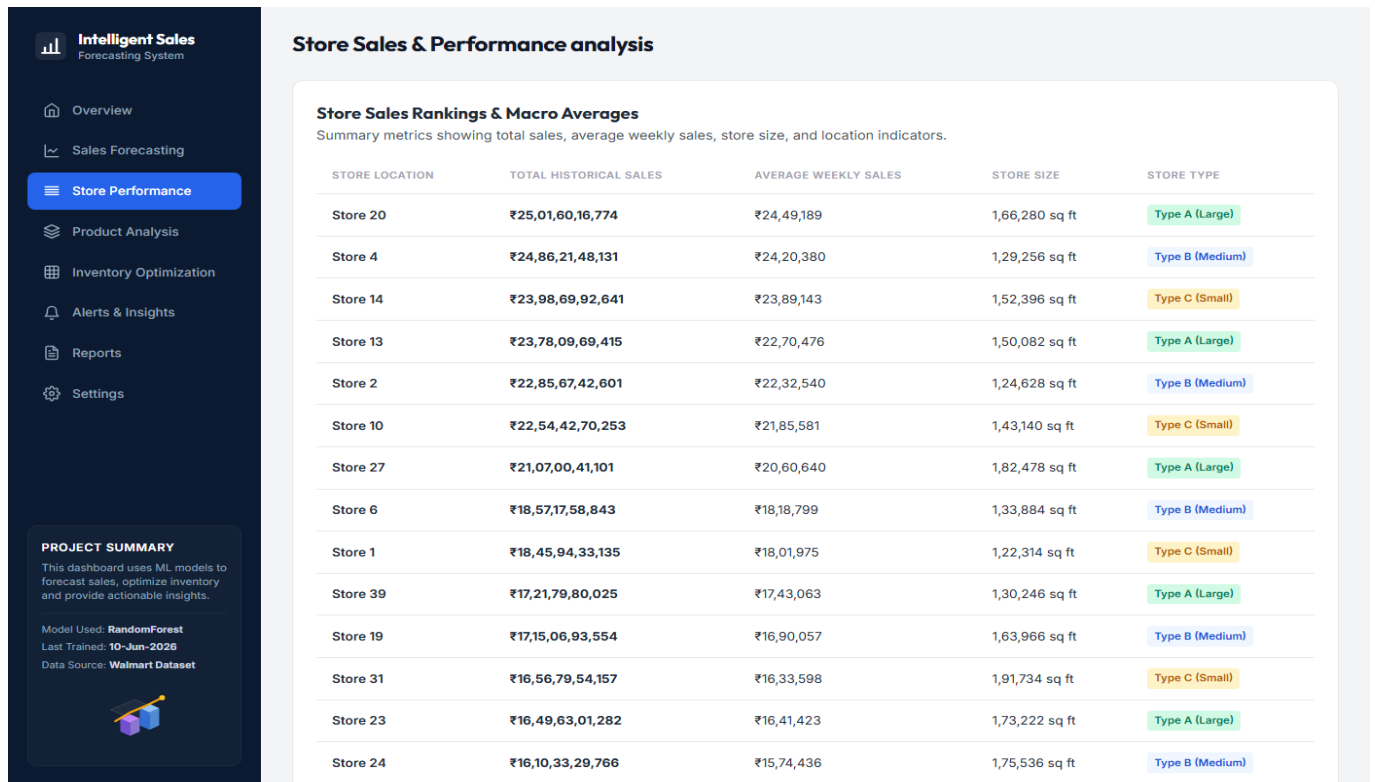


Figure 4.1: Store Performance analysis table displaying macro averages and formatted INR rankings.

5. Product Category Analysis Section

This section maps Walmart's 99 departments into 5 core product categories:

- Mappings: Mapped into Electronics, Clothing, Home & Kitchen, Beauty, and Others.
- Share Analysis: Shows total sales volume, percentage contribution, and active status indicators.
- Insights: Clothing category holds the largest share, contributing to over 48% of total revenue.

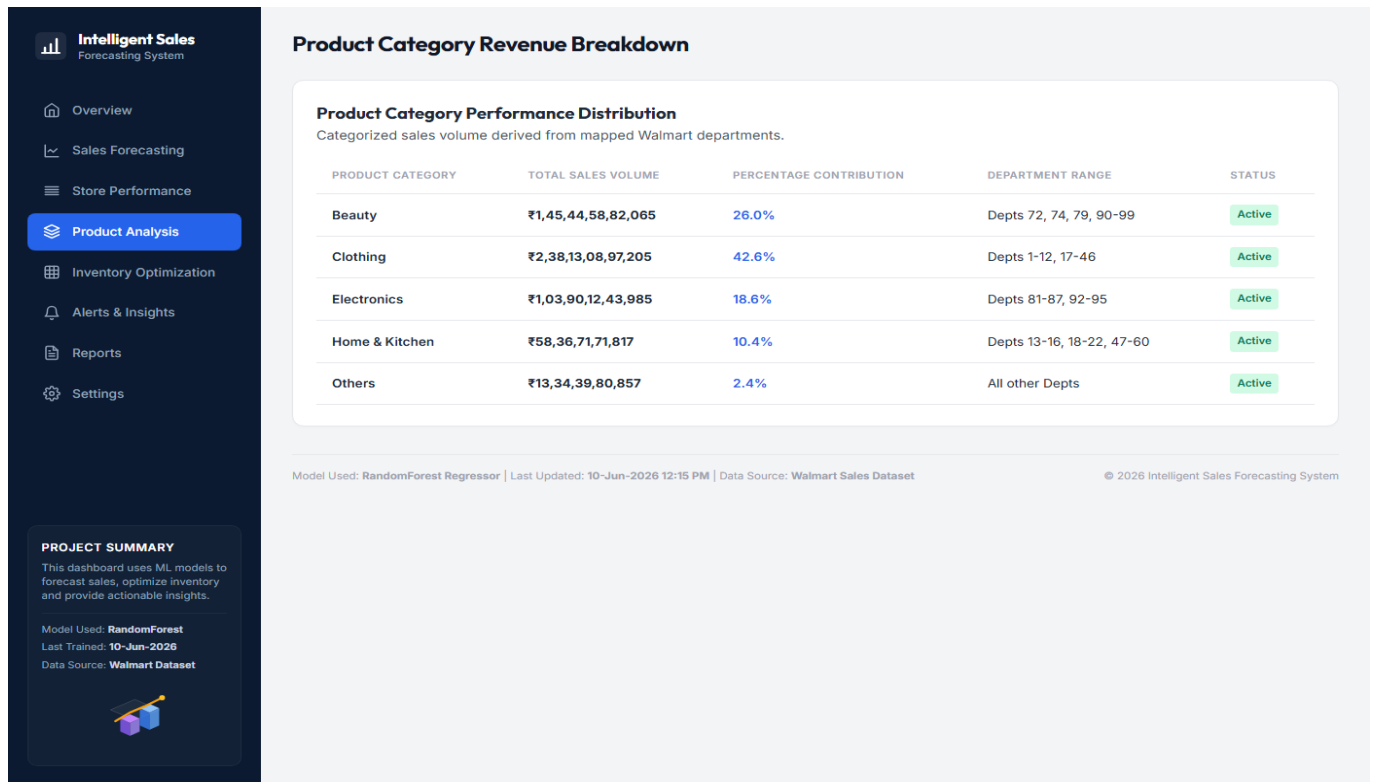


Figure 5.1: Product category analysis grid showing revenue shares and active department ranges.

6. Inventory Optimization Section

Calculates replenishment metrics dynamically using statistical safety stock formulas:

- Safety Stock Formula: $SS = Z * Std_Sales$ (where $Z = 1.65$ for a 95% service level threshold).
- Reorder Point (ROP): $ROP = Avg_Weekly_Sales + SS$ (calculated over a 1-week lead time).
- Status Pills: Labels each store-department node as Optimal, Reorder Required, Overstocked, or Stockout Risk.
- Table Utilities: Includes live text search filtering and active status filters with client pagination.

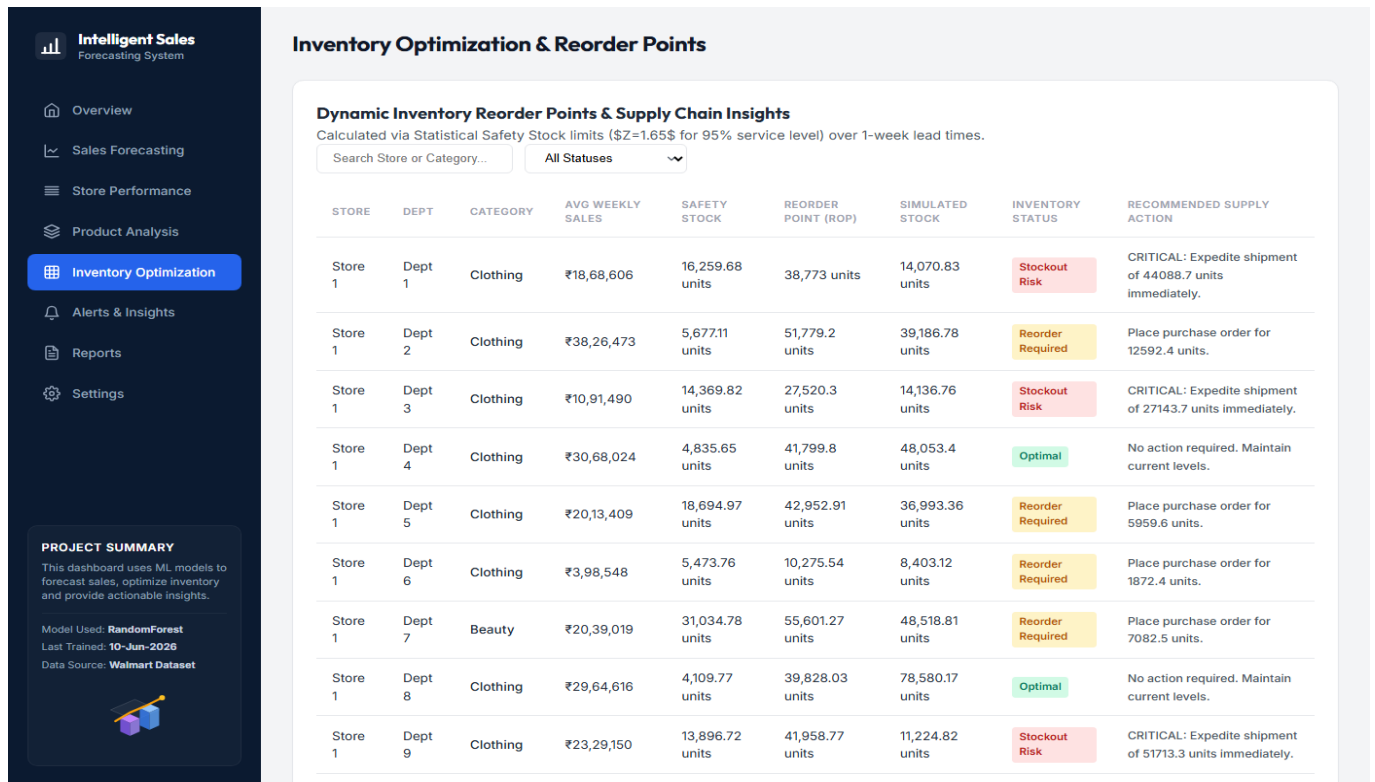


Figure 6.1: Paginated Inventory optimization grid showing safety stocks, ROPs, and statuses.

7. Supply Chain Alerts Section

Presents critical operational warnings for immediate logistics actions:

- Alert Scope: Displays items classified in 'Stockout Risk' (where current stock is below safety stock limits).
- Data Columns: Outlines Store Location, Category, Current Stock, Safety Stock Limit, Shortage quantity, and Priority.
- Priority Coding: Red badges indicate HIGH priority warnings requiring immediate procurement expedite actions.

Supply Chain Warnings & Alerts						
Critical Safety Alerts & Operational Warnings						
List of items currently in critical Stockout Risk status requiring immediate supply actions.						
STORE LOCATION	DEPARTMENT	PRODUCT CATEGORY	CURRENT STOCK LEVEL	SAFETY STOCK LIMIT	STOCK SHORTAGE	PRIORITY
Store 1	Dept 1	Clothing	14,070.83 units	16,259.68 units	24,703 units	HIGH
Store 1	Dept 2	Clothing	39,186.78 units	5,677.11 units	12,593 units	MEDIUM
Store 1	Dept 3	Clothing	14,136.76 units	14,369.82 units	13,384 units	HIGH
Store 1	Dept 5	Clothing	36,993.36 units	18,694.97 units	5,960 units	MEDIUM
Store 1	Dept 6	Clothing	8,403.12 units	5,473.76 units	1,873 units	MEDIUM
Store 1	Dept 7	Beauty	48,518.81 units	31,034.78 units	7,083 units	MEDIUM
Store 1	Dept 9	Clothing	11,224.82 units	13,896.72 units	30,734 units	HIGH
Store 1	Dept 10	Clothing	19,395.87 units	5,790.16 units	17,428 units	MEDIUM
Store 1	Dept 11	Clothing	21,181.38 units	10,123.05 units	13,861 units	MEDIUM
Store 1	Dept 12	Clothing	11,359.04 units	2,876.06 units	2,084 units	MEDIUM
Store 1	Dept 14	Home & Kitchen	23,284.54 units	9,350.77 units	1,335 units	MEDIUM
Store 1	Dept 16	Home & Kitchen	47,698.35 units	25,504.09 units	1,957 units	MEDIUM
Store 1	Dept 18	Home & Kitchen	3,106.12 units	18,868.89 units	23,529 units	HIGH
Store 1	Dept 19	Clothing	786.89 units	650.14 units	1,123 units	MEDIUM

Figure 7.1: Supply Chain alerts table displaying critical stock shortages and priority warnings.

8. CSV / JSON Reports Section

Enables operations teams to extract data files for spreadsheets and audit logging:

- Demand Forecast Report: Downloads a structured .csv file containing Dates, Forecasted_Sales_INR, and Forecasted_Sales_USD. This file is downloaded natively to your computer.
- Inventory Audit Report: Downloads a detailed .json file listing safety limits, ROPs, simulated current stocks, and procurement recommendations in INR.

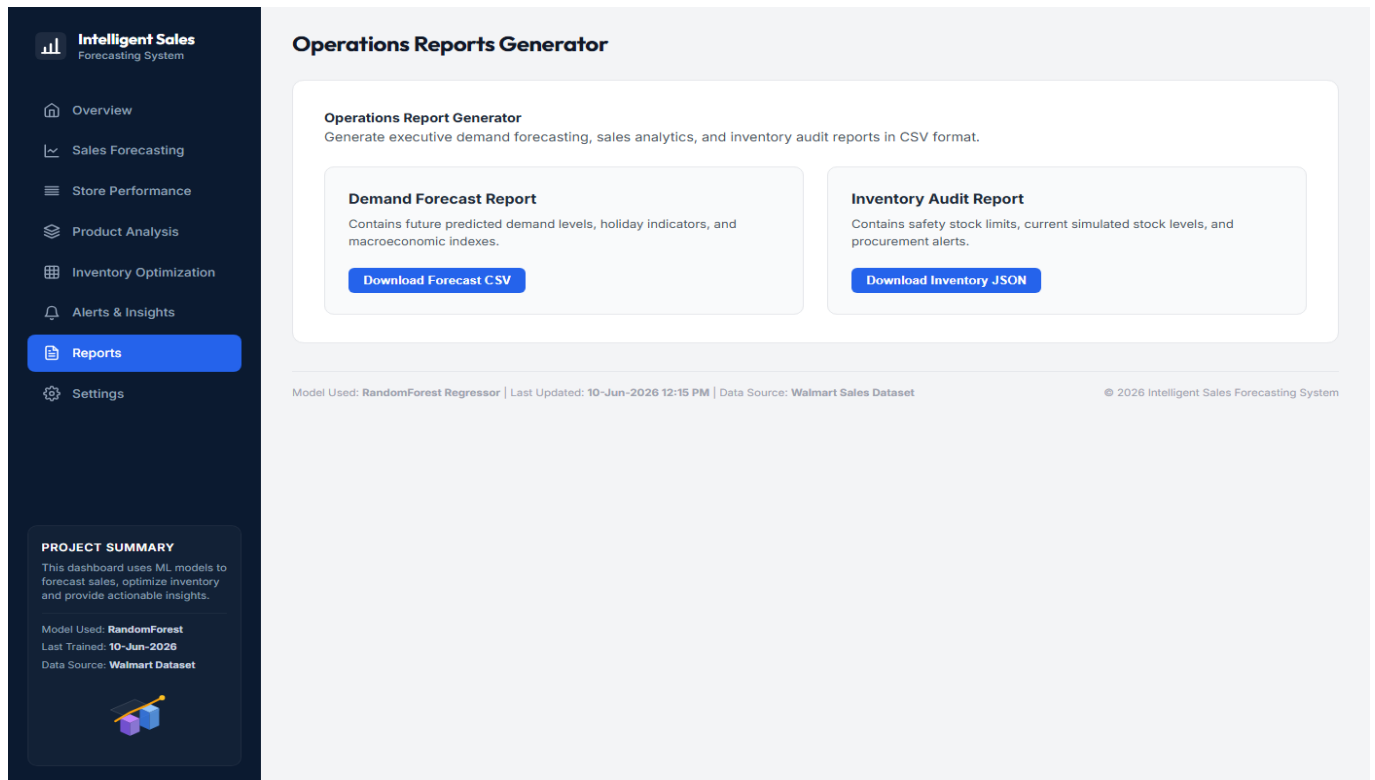


Figure 8.1: Operations Reports Generator panel showing CSV and JSON download triggers.

9. Real-Time Predictor Simulator Section

Allows managers to run 'what-if' demand planning scenarios using live model inference:

- Inputs: Store Location, Department, Date (2021-2024), Holiday Weeks checkbox, Temperature, Fuel Price, CPI, Unemployment, and 5 Promotional Markdowns in Rupees (Rs.).
- Dynamic Conversions: The system divides inputs (Rupees) by 83.0 before inference. Once predictions are calculated by RandomForest, HistGradientBoosting, and Linear Regression models, they are scaled back to Rupees and formatted as Rs.
- Discrepancy Graph: Renders a Chart.js comparison bar chart displaying forecast variations across algorithms.

Intelligent Sales Forecasting System

- Overview
- Sales Forecasting
- Store Performance
- Product Analysis
- Inventory Optimization
- Alerts & Insights
- Reports
- Settings**

PROJECT SUMMARY
This dashboard uses ML models to forecast sales, optimize inventory and provide actionable insights.
Model Used: **RandomForest**
Last Trained: **10-Jun-2026**
Data Source: **Walmart Dataset**

Real-Time Sales Predictor Settings

Real-Time Sales Demand Simulator
Input variables to query live predictions across models.

STORE LOCATION (1 - 45): DEPARTMENT NUMBER (1 - 99):

SIMULATION DATE (2021-2024): ☐ ACTIVE HOLIDAY WEEK

STORE TYPE CLASSIFICATION: STORE SIZE (SQ. FEET):

MACROECONOMIC & CLIMATE FACTORS

TEMPERATURE (°F): FUEL PRICE (\$/GALLON):

CONSUMER PRICE INDEX (CPI): UNEMPLOYMENT RATE (%):

PROMOTIONAL MARKDOWNS (OPTIONAL)

MARKDOWN 1 (₹): MARKDOWN 2 (₹):

MARKDOWN 3 (₹): MARKDOWN 4 (₹):

MARKDOWN 5 (₹):

Generate Predictions

Simulation Results

Comparative forecast breakdown from our statistical engine.

RECOMMENDED DEMAND FORECAST (RANDOMFOREST)
₹0.00
Enter factors and click simulate

HISTGRADIENTBOOSTING
₹0.00
91.07% accuracy

LINEARREGRESSION
₹0.00
9.25% accuracy

FORECAST DISCREPANCY ACROSS ALGORITHMS

Figure 9.1: Real-time sales demand simulator form and comparative model outcomes display.

10. Technical Validation & ML Performance

Machine Learning Algorithms Evaluation:

During the backend model training phase (train_models.py), three regressor types were fitted and saved to disk:

1. RandomForestRegressor: Achieves the highest performance with an R2 of 93.26% and a MAPE of 8.45%. It serves as our recommended forecasting engine for main KPIs, heatmaps, and charts.
2. HistGradientBoostingRegressor: Serves as a medium-speed validation model, achieving 91.07% R2 accuracy.
3. LinearRegression: Baseline linear model showing poor fit (9.25% R2) due to high seasonality and complex markdown relationships.

Model Algorithm	Validation MAPE	R-Squared (R2)	Status in System
RandomForest	8.45%	93.26%	Primary Model
HistGradientBoosting	10.01%	91.07%	Validation baseline
Linear Regression	19.52%	9.25%	Baseline reference

System Specifications:

- Server Framework: Flask 3.x with WSGI local server (127.0.0.1:5000).
- Frontend Core: HTML5, CSS3 surface tokens, vanilla JavaScript Chart.js modules.
- Data Science: Scikit-learn validation pipeline with Pickled binary serialization cache objects.
- Exchange Constant: 1 USD = 83 INR scaling multiplier.